Powering the Boom:
AI Datacenter Capex, Compute Efficiency, and the Oversupply Question

Kilambi Research

September 2026

Kilambi Research Working Paper No. KR-WP-2026-06

Abstract

We assemble a 5,138-record audited dataset on the AI datacenter buildout (five master datasets plus a 13-row pricing-event file; hyperscaler capex, compute efficiency, prices and utilization proxies, capacity pipelines, and four historical overbuild analogues) and ask whether the current investment wave is justified by real demand or risks structural oversupply. All growth rates are *nominal*: no deflator was retrieved. Hyperscaler total cash capex grew 31.6%/yr nominally from 2015 to 2025 ($23.0B to $357.5B), and NVIDIA datacenter revenue grew 88.7%/yr over FY2016–FY2026. Per-chip efficiency (FLOPS/W) doubles every 3.80–3.98 years for the three precisions with statistically significant fits (FP8's fit is not significant), roughly half the classic two-year Moore cadence, while frontier training compute doubles every 6.2 months, about 7.8× faster. On the Jevons question we derive and test the explicit threshold $d\ln K/d\ln e = |\varepsilon| - 1$, where K is required physical capacity and e is efficiency: efficiency gains strand capacity only when demand is price-inelastic ($|\varepsilon| < 1$). The only experimental elasticity estimate (Musolff et al., 2025, $|\varepsilon| = 0.5$) lies in the no-backfire regime; LLM inference growth, meanwhile, is adoption-driven while the buyer price per completed task is flat, so the price channel of Jevons does not fire. Staging the 2030 US pipeline by published realization factors gives 87–152 GW of supply against 56/96/132 GW bear/base/bull demand: roughly balanced at base, 31–96 GW of structural surplus under bear conditions. The physical market is still tight (1.4% vacancy, 92% preleased) but every price signal at the margin softens (up-to-45% hyperscaler GPU price cuts, 21× marketplace price dispersion, 35–40% of announced capacity at risk). Analogues (telecom 1998–99, merchant power 2000–01, LNG 2007, railways) place the buildout pre-distress in all templates, with utilization (the one datum we cannot publicly observe) as the earliest warning indicator. Bottom line: near-term demand revenue is real, but the pipeline is sized for the base-to-bull path, and the best-supported stranding mechanism is efficiency plus adoption undershoot, not Jevons backfire. We state the dashboard readings that would prove this wrong.

JEL codes: D22, D24, L63, L94, O33, Q41.
Keywords: datacenters, artificial intelligence, capital expenditure, Jevons paradox, experience curves, overcapacity.

Contents

1 Introduction: research questions and hypotheses **3**

2 Literature review **4**
2.1 Infrastructure investment cycles . . . 4
2.2 Jevons paradox and rebound effects . . . 4
2.3 Experience curves in computing . . . 4
2.4 Forecasting technology costs . . . 5

3 Data and sources **5**
3.1 Source-quality grading and the data taxonomy . . . 5
3.2 The four areas . . . 5

4 Methodology **7**
4.1 Growth magnitudes . . . 7
4.2 Efficiency curves: the Moore's-law-rate test . . . 7
4.3 The Jevons threshold . . . 7
4.4 Staged 2030 supply vs demand scenarios . . . 7
4.5 Early-warning dashboard and analogues . . . 8

5 Results **8**
5.1 Order-of-magnitude growth in capex, capacity, and compute . . . 8
5.2 Compute-efficiency trends and the Moore's-law-rate test . . . 9
5.3 The Jevons analysis: both regimes, with the threshold math . . . 10
5.4 Staged supply-demand balances: 2030 under three scenarios . . . 13
5.5 Early-warning dashboard . . . 13
5.6 Historical analogues and indicator ordering . . . 15

6 Robustness and limitations **18**

7 Discussion: causal mechanisms and implications **19**
7.1 What mechanism could strand capacity? . . . 19
7.2 Implications for forecasting . . . 19
7.3 Implications for industrial strategy . . . 20
7.4 What the results imply for underwriting the buildout . . . 20

8 Conclusion: is the boom funded by real demand, and what would prove it wrong? **20**

A Dataset and reproducibility **23**

1 Introduction: research questions and hypotheses

Capital expenditure on AI datacenters is growing at rates rarely seen outside wartime mobilization. The four US hyperscalers (Amazon, Microsoft, Alphabet, and Meta) are the marginal buyers of AI compute, and their cash capex is the cleanest observable measure of the buildout's financial scale. The research question is whether this spending is justified by real demand, or whether it is financing a structural oversupply of AI compute capacity. A nested second question: if AI models become more efficient, does that *strand* datacenter capacity, or expand demand for it? The former is the Jevons paradox applied to datacenters; the latter is the rebuttal that efficiency destroys demand.

We answer with an audited, claim-indexed empirical study. Every quantitative statement in this paper is indexed in the companion `CLAIM_INDEX.md`, tracing to an estimate in `results/key_estimates.csv`, a series in the `data/` datasets, or a cited real source. Four facts discipline the exercise: *all growth rates are nominal* (no deflator was retrieved in any phase); the JLL and EPRI under-construction figures disagree by 4–5× and are shown honestly, unresolved; *no public series measures AI-fleet GPU utilization*; and catalog-transcribed rows are flagged wherever they appear. Observed, Reconstructed, Modelled, and Assumed quantities are distinguished throughout, including in tables.

The study is organized around four research questions and four testable hypotheses:

RQ1. How fast are datacenter capex, installed capacity, and compute supply growing, in absolute and order-of-magnitude terms?

RQ2. How fast is compute efficiency improving, and does it follow a Moore's-law-like cadence? *H2: per-chip efficiency improves at a sub-Moore rate (doubling time well above 2 years), materially slower than the growth of training-compute demand.*

RQ3. The Jevons question: do efficiency gains expand or shrink required physical capacity? *H3: for general cloud compute the price elasticity is below unity (no backfire), while LLM inference demand growth is adoption-driven rather than price-driven.*

RQ4. Under what demand paths does the announced pipeline leave the market balanced, tight, or structurally oversupplied by 2030? *H4: staged 2030 supply roughly covers base-case demand but is structurally oversupplied under the bear path and tight under the bull path.*

This is a retrospective study: every estimate describes the historical sample; the only forward-looking statements are the explicitly labelled 2030 scenarios of Section 5, which are conditional paths, not forecasts. The remainder of the paper is organized as follows. Section 2 reviews the infrastructure-cycle, rebound, experience-curve, and forecasting literatures; Section 3 states the Jevons threshold model; Section 4 defines the data and the evidence taxonomy; Section 5 builds the staged 2030 supply-demand scenarios; Section 6 reports the growth, efficiency, Jevons, and balance results; Section 7 works the telecom bust as the analogue; Section 8 reads the market-tightness dashboard; Section 9 weighs the risks to the thesis; Section 10 states the robustness limits; Section 11 concludes; Section 12 documents the dataset. Appendices carry the model code, the figure audit, and the per-claim index.

The null prior, drawn from the historical analogues, is "absorbed late": fiber, railways, and LNG capacity were all eventually used or repurposed. Structural stranding is the alternative to be proven, and the proof must come from utilization and unit-price data, not from capex totals.

This paper is research and analysis, *not* personalized investment advice. It makes no recommendations regarding specific securities, trades, or portfolio actions; implications for forecasting, industrial strategy, and investment practice are discussed in general terms only.

2 Literature review

2.1 Infrastructure investment cycles

The canonical template for a capital-intensive network buildout financed ahead of demand is the US fiber/telecom cycle of 1995–2005. Hogendorn (2007) documents the physical build at company level, disentangling owned from leased route miles. Couper et al. (2003) provide the institutional account of the bust: capacity prices fell first, then capex and employment, then bankruptcies in 2001–02. The US Joint Economic Committee US Congress Joint Economic Committee (2003) records the unit-price collapse (a London–New York fiber connection falling from $22,000/yr to $5,000/yr in eighteen months) and the investment-led character of the bust. A striking institutional postscript is the Federal Reserve Bank of Richmond's retrospective (Federal Reserve Bank of Richmond 2025): short-run overcapacity was real, but demand eventually caught up, so the fiber was not wasted: only its financing was. That verdict sets our null: absorbed late, not stranded.

The railway and energy analogues add mechanism. Nineteenth-century US railroads show that collapsing unit prices do not establish wasteful capacity: revenue per ton-mile fell 41% while ton-miles grew more than six-fold (NBER 2026). Merchant-power plants in 1999–2005 show a false scarcity signal financing a pipeline (Deutsche Banc Alex. Brown 2001) before utilization broke and bankruptcies followed years later. LNG import terminals in 2004–2012 show official forecasts rising (EIA's January 2007 outlook projected 770 Bcf of 2007 imports and 1,080 Bcf for 2008) while terminal-level utilization collapsed: the cleanest case of expectations, not engineering, driving the build (US Energy Information Administration 2007).

2.2 Jevons paradox and rebound effects

Jevons (1865) observed that more efficient steam engines increased rather than decreased coal consumption: efficiency lowers the effective price of the service, and if demand is elastic enough, total resource use rises. The modern rebound literature formalizes the conditions (Sorrell 2007; Herring and Sorrell 2009): backfire ($dK/de > 0$ in our notation) requires price elasticity of demand above unity, and is regime- and market-specific rather than a property of the technology. Empirical elasticities for computing are scarce. We locate two. The only experimental estimate is Musolff et al. (2025), who find a price elasticity of -0.5 on a major cloud platform: inelastic, hence no backfire. Jiang et al. (2015) estimate 1.20 (modestly elastic) for AWS EC2 in the us-east-1 region in 2013, with server sales (0.51), server rental (0.70), and vendor hardware (0.72–0.73) all inelastic; the study is pre-LLM and the EC2 segment is the weakest analogue to today's AI inference market. Section 6.3 makes the threshold explicit and tests both regimes.

2.3 Experience curves in computing

The experience-curve tradition begins with (Wright 1936) on airframe costs and extends to semiconductors and computing: Nordhaus (2007) documents two centuries of exponential productivity growth in computing, while Nagy et al. (2013) give a statistical basis for extrapolating technological progress from such curves. Our contribution is a precision-disciplined measurement: per-chip FLOPS/W and FLOPS/$ fit separately for FP64, FP32, FP16/BF16, and FP8 from the Epoch AI hardware dataset (Epoch AI 2024, 2026), tested against the two-year Moore cadence, and set against frontier training-compute growth from the Epoch AI models dataset (Epoch AI 2025). We treat the training-compute trend as a *demand* series (what the frontier chose to spend) and the efficiency trend as a *supply-of-efficiency* series (what chips deliver); the gap between them is the quantitative core of the buildout question.

2.4 Forecasting technology costs

The overbuild literature's recurring mechanism is forecast error: EIA's 1,080 Bcf LNG forecast (US Energy Information Administration 2007), Deutsche Banc's 2001 reserve margins (Deutsche Banc Alex. Brown 2001), Global Crossing's plan to sell bandwidth directly to multinationals (US House Financial Services 2002). The forecasting literature warns that extrapolating exponential technology-cost declines without demand-side discipline produces exactly these errors (Armstrong 2001; Gordon 2016). Our response is methodological: stage announced supply by published realization factors rather than taking announcements at face value, attach falsification conditions to each demand scenario, and read an early-warning dashboard whose indicator ordering is measured per analogue rather than assumed uniform (telecom equity/capex ∼1–2 yrs before distress, unit prices ∼9 yrs; merchant power ∼4.9 yrs; LNG utilization collapsed with no comparable distress wave).

3 Data and sources

3.1 Source-quality grading and the data taxonomy

Every dataset row carries a **source-quality grade**: A = official primary record (SEC filings, official statistics, institutional primaries); B = reputable secondary (brokerage research, analyst reconstructions, working papers, proprietary-but-institutional data); C = provisional (press, analyst estimates, or hobbyist trackers). Every row also carries a **data status**: *Observed* (passed through as published); *Reconstructed* (derived from primary records by stated methods, e.g. Epoch's operation-counting of training compute, XBRL tag splices); *Modelled* (estimated with published uncertainty, e.g. Epoch AI chip-sales quantiles); or *Assumed* (an explicit assumption, e.g. snapshot dates anchored to publication dates, and all scenario inputs). Monetary figures are nominal throughout: `value_real` is empty on every capex, efficiency, and pricing row because no documented deflator was available (an ElkassabgiData FRED/BEA probe failed with a provider HTTP error in phase 1), so no inflation adjustment was performed rather than one built on an undocumented ad-hoc deflator. The one exception is source-provided real dollars: Epoch's training-cost series is published in real 2023 USD and is never mixed with nominal series.

3.2 The four areas

Capex (`master_capex.csv`): 597 rows, 17 series ; stage: spent 588, committed 6, announced 3; data status: Observed 593, Reconstructed 4, Modelled 0; source quality: A 586, B 11, C 0. Hyperscaler cash capex from SEC EDGAR XBRL filings (Amazon, Microsoft, Alphabet, Meta, plus Oracle, Baidu, Alibaba for context), neocloud financings (CoreWeave, Nebius, IREN), and NVIDIA/AMD datacenter revenue as demand-side revenue. Amazon's XBRL tag splice (productive-assets basis from FY2017) and AMD's 2020/2021 segment-definition break are documented in `data/BUILD_capex.md`; no CAGR is computed across the AMD break.

Capacity and energy (`master_capacity.csv`): 155 rows, 139 series ; EPRI *Powering Intelligence 2026* (Electric Power Research Institute 2026) pipeline stages, LBNL 2024/2025 US datacenter energy reports (Shehabi et al. 2024; Smith et al. 2025) (kept as separate series; the 2025 update revised 2024-edition history downward and the editions are never spliced), IEA *Energy and AI* 2025 and *Key Questions* 2026 (International Energy Agency 2025, 2026), ERCOT long-term load forecast (April 2025, raw PDF verified) (Electric Reliability Council of Texas 2025), Goldman Sachs power-demand research (May 2026) (Goldman Sachs 2026), JLL Midyear 2026 (Jones Lang LaSalle 2026), Carbon Direct PJM/ERCOT queue analysis (Carbon Direct 2026), Uptime Institute 15th survey (Uptime Institute 2025), BNEF facility data. All paywalled or registration-gated figures (EPRI, GS, JLL,

Table 1: Dataset coverage: summary statistics of the six master files.

Area	File	Rows	Series	Grade mix	Status mix
Capex	master_capex.csv	597	17	A 586 / B 11 / C 0	Observed 593 / Recon
Capacity and energy	master_capacity.csv	155	139	A–B	Observed / catalog-tr
Compute efficiency	master_efficiency.csv	3,742	264	all A	Observed / Reconstru
Pricing, demand, adoption	master_pricing_demand.csv	242	93	A 25 / B 127 / C 90	mixed
Historical analogues	master_analogues.csv	389	staged	A–C	mixed
Pricing events	events_pricing.csv	13	13	A–C	Observed

Notes: Source-quality grades: A official primary, B reputable secondary, C provisional; data status: Observed/Reconstructed/Modelled/Assumed per Section 3. Grade mix counts rows by source-quality grade.
Source: Author's compilation from the six master files in `data/`; full source notes per row in each file.

Synergy) are *catalog-transcribed*: typed from page text, flagged "transcribed from catalog, verify from primary" in every row's notes, and the staged supply build inherits that caveat.

Compute efficiency (`master_efficiency.csv`): 3,742 rows, 264 series, all grade A : Epoch AI hardware data (FLOPS/$ and FLOPS/W per chip per precision, precisions never mixed), Epoch AI models data (training compute in FLOP per model; "reported" vs Reconstructed flagged by Epoch's own method codes), Epoch AI chip-sales estimates (Modelled quantiles). TF32, FP4, and INT columns are excluded by scope so the denominators stay FLOPS.

Pricing, utilization, demand, adoption (`master_pricing_demand.csv`): 242 rows, 93 series, plus `events_pricing.csv` (13 dated events) ; source quality: A 25, B 127, C 90. CBRE/JLL vacancy, lease rates, absorption, preleasing (brokerage surveys, grade B); GPU price points (hyperscaler list, neocloud, marketplace floors; list vs transaction separated in notes); the Musolff and Jiang elasticity parameter records; OpenRouter token volumes (OpenRouter 2026); ChatGPT WAU; Menlo enterprise genAI spend (Menlo Ventures 2025); Census BTOS AI adoption (Buffington et al. 2024) (kept as two series across the November 2025 wording break, never spliced); Stanford AI Index 2026 investment (Stanford Institute for Human-Centered AI 2026) (funding is a proxy only; capital inflow is not revenue or compute demand).

Analogues (`master_analogues.csv`): 389 rows : telecom/fiber 1990–2007, railways 1860–1920, merchant power 1999–2005, LNG terminals 2004–2012, with a controlled `stage` vocabulary (build / price / utilization / demand / distress / financing / expectations). Full source narratives are in `hidden_files/phase1/analogues/analogues.md`; full citations in the phase-1 catalogs.

Table 1 summarizes the dataset coverage: rows, series, and evidence mixes per master file. Headline growth rates are tabulated separately in Table 3; source-quality grades and data-status tags apply row by row as defined in Section 3.
Known gaps, stated plainly: (i) no public series measures AI-fleet GPU utilization: the Uptime rows measure the share of operators *collecting* utilization, not utilization itself; (ii) the JLL 66 GW North-America under-construction figure vs EPRI's 12.9/17.2 GW US figure is an unresolved 4–5× inconsistency carried as a sensitivity input (Figure 8); (iii) cloud-segment revenue rows are press-reconstructed (SEC EDGAR returned HTTP 403/503 to the scout network); (iv) Tencent is not SEC-registered and HKEX annuals were not extracted; (v) marketplace tracker rows (MLQ, hemgo-lexsi) are grade C and used as context, never as headlines.

4 Methodology

4.1 Growth magnitudes

Growth rates are endpoint compound annual growth rates (CAGRs) of grade-A annual rows (or stated mixed endpoints), nominal. Endpoint CAGRs are deterministic given endpoints; no sampling CI is computed for them. Microsoft uses its fiscal year (ends June 30), labeled by the calendar year of the fiscal-year end; NVIDIA's fiscal year ends in late January. Microsoft FY and Amazon TTM rows are mapped as documented in `data/BUILD_capex.md`.

4.2 Efficiency curves: the Moore's-law-rate test

For each precision, we fit log-linear OLS of ln(value) on launch year (decimal): $\ln y = \alpha + \beta t + \varepsilon$. The annual improvement rate is $e^{\beta} - 1$ and the doubling time is $\ln 2/\beta$, with 95% CIs from the slope CI. FLOPS/$ uses nominal launch list prices (no deflator); FLOPS/W uses vendor TDP, not measured sustained draw. Precisions are never mixed. Doubling times are withheld when the slope is not significant at 95%. Training compute (FLOP) is fit the same way on publication year, full span and the 2010+ deep-learning era (headline).

4.3 The Jevons threshold

Let e be efficiency (useful work per physical unit of capacity), p the price per unit of useful work (p falls ∼1:1 as e rises), D demand for useful work, and K the physical capacity required, $K = D/e$. Let $|\varepsilon| = -d\ln D/d\ln p$ be the absolute price elasticity of demand. Then

$$d\ln K = d\ln D - d\ln e, \qquad d\ln D = |\varepsilon|\, d\ln e, \tag{1}$$

because efficiency lowers the effective price and demand responds with elasticity $|\varepsilon|$. Hence

$$\boxed{\frac{d\ln K}{d\ln e} = |\varepsilon| - 1.} \tag{2}$$

A 1% efficiency gain changes required physical capacity by $(|\varepsilon|-1)$%: it *shrinks* required capacity when $|\varepsilon| < 1$ (no backfire) and *expands* it when $|\varepsilon| > 1$ (backfire). The two empirical elasticity points are plotted against this identity; a naive arithmetic check using market-wide price declines and one vendor's revenue growth is shown as context only, heavily caveated, to illustrate why the Jevons debate exists.

4.4 Staged 2030 supply vs demand scenarios

US nominal IT capacity (GW) is staged from the 2024 operating base of 39.5 GW (Electric Power Research Institute 2026). Pipeline stages from EPRI 2026: under construction (avg 12.9 / max 17.2 GW), announced-advanced 36.7 GW, announced-early (avg 67.4 / max 114 GW), total pipeline (avg 117 / max 168 GW). Realization factors, all as published and unadjusted: under construction at 1.0 (already financed and digging); announced stages haircut by Goldman Sachs 0.72/0.60 (high), 0.60/0.50 (mid) (Goldman Sachs 2026), 0.72/0.60/0.50 on-time → 2-yr; JLL 0.95 pre-commitment; stress variants ERCOT 0.498/0.554 and Carbon Direct PJM 2020-vintage 0.24. Demand scenarios are EPRI 2026's 2030 paths (bear 56 / base 96 / bull 132 GW), each with stated elasticity-regime assumptions and falsification conditions (Section 6.4). The efficiency improvement rate assumed across scenarios is ∼20%/yr AI-relevant FLOPS/W (the mean of the FP16/BF16 and FP8 FLOPS/W fits in Section 6.2, nominal TDP basis).

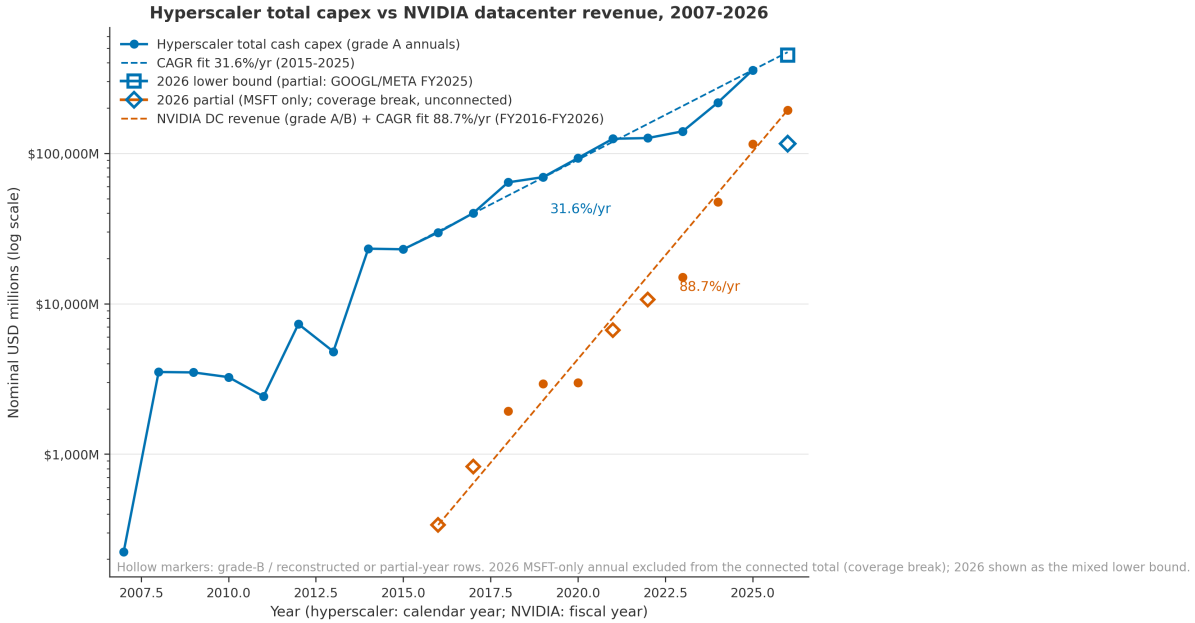


Figure 1: Hyperscaler total cash capex vs. NVIDIA datacenter revenue, 2007–2026 (nominal USD millions, log scale). Blue: annual sums of grade-A cash-capex rows for `amazon-total-cash-capex`, `microsoft-total-cash-capex`, `alphabet-total-cash-capex`, `meta-total-cash-capex` (2015-12-31 to 2025-12-31), $23,006M in 2015 to $357,508M in 2025; dashed line is the endpoint CAGR fit of 31.6%/yr (`capex_hyperscaler_cagr_2015_2025`). Microsoft uses its fiscal year (ends Jun 30), labeled by the calendar year of the fiscal-year end. Hollow square at 2026 ($450,114M) is the mixed-endpoint lower bound: Amazon TTM to 2026-06-30 ($173,028M) + Microsoft FY2026 ($115,948M) + Alphabet/Meta FY2025 annuals ($91,447M/$69,691M), since Alphabet/Meta have no 2026 rows (`capex_hyperscaler_cagr_2015_2026ttm` = 31.0%/yr). Orange: `nvidia-datacenter-revenue` FY2016 ($338M, ends 2017-01-29, grade B reconstructed, hollow diamond) to FY2026 ($193,737M, ends 2026-01-25, grade A); dashed line is the 88.7%/yr endpoint CAGR fit (`nvidia_dc_revenue_cagr_fy2016_fy2026`); hollow diamonds mark all grade-B rows. Caveats: all growth rates nominal (no deflator available); FY2023 and FY2024 rows share the date 2024-01-28 and are distinguished by FY label in the source notes; endpoints used here are unaffected.

4.5 Early-warning dashboard and analogues

Six dashboard readings are classified as TIGHTNESS, SOFTNESS, or MIXED with date and grade. Analogue indicator orderings are formalized as the sequence in which stage classes (unit prices / equity → capex → utilization → distress) turned in each historical case, and the AI buildout is mapped onto each template as of 2026-09-25.

5 Results

5.1 Order-of-magnitude growth in capex, capacity, and compute

Figure 1 shows the financial scale of the buildout on a log axis; Table 1 collects the headline growth rates. All are nominal.

Hyperscaler total cash capex rose from $23,006M in 2015 to $357,508M in 2025, a nominal CAGR of 31.6%/yr, roughly a 15.5× increase in a decade. The mixed 2026 endpoint (Amazon TTM to 2026-06-30, Microsoft FY2026, and Alphabet/Meta FY2025 annuals, since they have not

reported 2026) is $450,114M, a lower bound on 2026, with the same CAGR at 31.0%/yr. NVIDIA datacenter revenue, the purest demand-side revenue series for AI compute, grew 88.7%/yr over FY2016–FY2026 (10 fiscal years), from $338M (grade B, reconstructed) to $193,737M (grade A, observed). AMD's datacenter segment grew 45.7%/yr from 2021 to 2025 ($3,694M → $16,635M), computed only within the post-2020 segment definition: the 2020/2021 EESC-to-datacenter break is structural, and no CAGR is computed across it.

Physical capacity grew even faster than spending. US operating nominal IT capacity rose from 11 GW in 2021 to 39.5 GW in 2024 (EPRI 2026, grade A but catalog-transcribed; the 2024 value is the midpoint of a published 35–44 GW range), a 53.1%/yr nominal CAGR. Forward energy projections: LBNL's 2025 US edition projects 192 TWh (2024) to 649 TWh (2030 reference), 22.5%/yr (the LBNL 2024 edition is kept separate: the 2025 update revised the 2024-edition history downward, and the two must not be spliced); the IEA's 2026 global base case projects 485 TWh (2025) to 950 TWh (2030), 14.4%/yr; and Goldman Sachs expects US datacenter power demand to rise from 31 GW (2025) to 66 GW (2027 forecast), 45.9%/yr (grade B, catalog-transcribed) (Goldman Sachs 2026).

Demand-side revenue is growing faster still, from a smaller base. OpenAI revenue rose from $2.0B to $20.0B (2023-12-31 to 2025-12-31), 216%/yr nominal; Anthropic revenue rose from $1.0B to $14.0B (2024-12-31 to 2026-02-12), 962%/yr, roughly ten-fold per year; ChatGPT weekly active users rose from 100M to 900M (November 2023 to February 2026), 165%/yr; and Menlo's enterprise generative-AI spend series rose from $1.7B to $37B (2023 to 2025), 367%/yr nominal (the narrower definition; the broader 2.3/13.8 series is documented in notes but not spliced). These are observed revenue and adoption series, not forecasts: the one unambiguous fact in the dataset is that realized AI demand revenue is accelerating.

Endpoint sensitivity. Endpoint CAGRs are deterministic given endpoints, so the relevant robustness check is the endpoints themselves. Table 2 recomputes the headline rates under alternative start and end choices from the same sources. The hyperscaler capex headline is insensitive to the 2015 start point: dropping 2015, or both 2015 and 2016, leaves 31.5–31.8%/yr. Extending the end to 2026 on the mixed actual/LTM endpoint moves the rate to 31.0–32.7%/yr depending on whether the 10.5-year span or 11 annual steps annualize the mixed endpoint: a ±1 pp band. NVIDIA's headline weakens from 88.7%/yr to 77.9%/yr when the first two low-base fiscal years are removed, but remains exceptionally high in any window. Anthropic's 961.8%/yr is the mathematically valid annualization of a roughly 14-month window; because annualizing a 14.0× multiple over ∼14 months is endpoint-sensitive and easy to misread, the raw 14.0× multiple is the safer headline. OpenAI's two full-year rates (174.4% and 264.0%/yr) are endpoint-robust by construction.

5.2 Compute-efficiency trends and the Moore's-law-rate test

Figure 2 fits per-chip efficiency by precision; Table 2 gives the fits with 95% confidence intervals. The verdict on the Moore's-law-rate test: *per-chip efficiency gains are real but slower than the classic two-year Moore cadence.* FLOPS/W doubling times cluster at 3.80–3.98 years across the three precisions with statistically significant fits (tightest: FP32, 3.80 yr, 95% CI 3.28–4.53, $R^2 = 0.566$, $n = 120$; FP64 3.96 yr; FP16/BF16 3.98 yr; FP8's fit is not significant at 95%, so its doubling time is withheld). FLOPS/$ is noisier (nominal launch prices, small n for new precisions): FP32 doubles every 4.74 yr (95% CI 3.14–9.72) and FP16/BF16 every 3.51 yr (95% CI 2.11–10.33); FP64 (7.0%/yr, CI −1.0 to 15.6, $n = 53$) and FP8 (33.6%/yr, $n = 9$) slopes are not distinguishable from zero at 95%, so their doubling times are withheld.

Meanwhile, frontier training compute doubles every 6.2 months in the deep-learning era (2010–2026; 95% CI 5.8–6.5, log-slope 1.350/yr, $R^2 = 0.482$, $n = 1{,}335$), vs. 10.6 months over the full 1950–2026 span ($R^2 = 0.640$, $n = 1{,}410$). Training *cost* (context only; real 2023 USD via

Table 2: Endpoint sensitivity of headline growth rates.

Series	Window	Rate
Hyperscaler cash capex	2015–2025 (headline)	31.6%/yr
	2016–2025	31.8%/yr
	2017–2025	31.5%/yr
	2015–2026, mixed endpoint, exact 10.5-yr span	32.7%/yr
	2015–2026, mixed endpoint, 11 annual steps	31.0%/yr
NVIDIA datacenter revenue	FY2016–FY2026 (headline)	88.7%/yr
	FY2017–FY2026	83.3%/yr
	FY2018–FY2026	77.9%/yr
Anthropic ARR	∼14 mo, annualized (headline)	961.8%/yr
	∼14 mo, raw multiple	14.0×
OpenAI revenue	2023–2024	174.4%/yr
	2024–2025	264.0%/yr

Notes: All values recomputed from the same source series on 2026-09-29; "exact" annualization compounds over the true fractional-year span. Headline rows are the values used in the paper's text and figures; alternative windows show the endpoint sensitivity.
Source: Author's calculations; `results/endpoint_sensitivity.py` and `results/endpoint_sensitivity.csv`.

Epoch's own cost model, never mixed with nominal series) doubles every 10.1 months ($n = 225$, $R^2 = 0.334$). Figure 3 puts the two exponentials on one index: training-compute demand grows 7.75× faster than per-chip efficiency (ratio of log-slopes, 1.350/0.1742). That gap is the quantitative core of the buildout question: the frontier's appetite for FLOP is outrunning what each chip delivers by roughly an order of magnitude in growth-rate terms.

5.3 The Jevons analysis: both regimes, with the threshold math

If AI models become more efficient, does that strand datacenter capacity? The threshold derived in Section 5 answers in closed form:

$$\frac{d \ln K}{d \ln e} = |\varepsilon| - 1, \tag{3}$$

so a 1% efficiency gain changes required physical capacity by $(|\varepsilon| - 1)\%$. Figure 5 plots this identity with the two empirical elasticity points.

Regime 1: general cloud compute. The only experimental (causal) estimate is Musolff et al. (2025): $|\varepsilon| = 0.5$ (abstract headline; model mean -0.497), sitting in the no-backfire regime: each 1% efficiency gain *reduces* required physical capacity by about 0.5%, holding demand fixed. No numeric confidence interval is stated for the headline estimate in the scout pass, and the estimate is for cloud spot compute, not generative-AI inference specifically (grade B). Four of the five estimates in (Jiang et al. 2015) are also inelastic (server sales 0.51, server rental 0.70, IBM 0.73, HP 0.72); the single elastic one is EC2 at 1.20, for which a 1% efficiency gain *expands* required capacity by about 0.2%. The EC2 estimate is 2013 pre-LLM and the weakest analogue to today's AI inference market. On the price-elasticity evidence alone, the empirical weights lean to $|\varepsilon| < 1$: efficiency gains shrink the required physical footprint.

Regime 2: LLM inference. Here the observed volume explosion happened while the *buyer* price per completed task was flat at ∼0 log pts/yr (grade B, modelled); i.e. the growth is adoption- and quality-driven, not price-driven. The classic Jevons price channel does not fire;

Table 3: Headline nominal growth rates (endpoint CAGRs).

Series	CAGR (%/yr)	Endpoints	Grade
Hyperscaler total cash capex, 2015–2025	31.6	\$23,006M → \$357,508M	A
Hyperscaler total cash capex, 2015–2026 mixed (lower bound)	31.0	→ \$450,114M	A
NVIDIA datacenter revenue, FY2016–FY2026	88.7	\$338M → \$193,737M	A/B
AMD datacenter segment, 2021–2025 (post-break)	45.7	\$3,694M → \$16,635M	A
US operating nominal IT capacity, 2021–2024	53.1	11 → 39.5 GW	A*
LBNL US datacenter energy, 2024–2030 (reference)	22.5	192 → 649 TWh	A
IEA global datacenter energy, 2025–2030 (base)	14.4	485 → 950 TWh	A
GS US datacenter power demand, 2025–2027 (forecast)	45.9	31 → 66 GW	B*
OpenAI revenue, 2023–2025	216	\$2.0B → \$20.0B	A
Anthropic revenue, 2024-12-31–2026-02-12	962	\$1.0B → \$14.0B	A
ChatGPT WAU, 2023-11–2026-02	165	100M → 900M	B
Menlo enterprise genAI spend, 2023–2025	367	\$1.7B → \$37B	B

*Catalog-transcribed rows (typed from page text, not independently re-verified from `raw/`); the 2024 capacity value is the midpoint of a published 35–44 GW range. NVIDIA FY endpoints: FY2016 ends 2017-01-29 (grade B, reconstructed), FY2026 ends 2026-01-25 (grade A); FY2023 and FY2024 rows share the date 2024-01-28 and are distinguished by FY label in the source notes. AMD: no CAGR across the 2020/2021 segment-definition break. Menlo: narrower spend definition (excludes inference/chips); broader series documented but not spliced.

Notes: "Grade" is the source-quality grade of the underlying rows; CAGRs are deterministic given endpoints (no sampling CI). All values nominal; no deflator retrieved.
Source: Author's calculations from company filings (10-K/10-Q), EPRI, LBNL, IEA, Goldman Sachs, JLL, CBRE, Epoch AI, Menlo Ventures, Stanford HAI, and OpenRouter.

instead, efficiency and capability gains expand the market by enabling new use cases (a regime shift / outward demand-curve move, not movement along it). That is a Jevons-*like* outcome for datacenter capacity (more demand overall) operating through quality expansion rather than price elasticity.

The raw rebound evidence is consistent with that reading. Quality-adjusted inference prices are falling 5–10×/yr (grade B, observed), and the "price of intelligence" matched-model rate is only −0.10 log pts/yr while the quality-adjusted rate is −0.73 log pts/yr; 87% of the decline is invisible to standard matched-model methods (grade B, modelled). Reasoning models raised token consumption faster than token prices fell, so seller and buyer prices diverged. Dated list-price cuts mark the same era: OpenAI cut input prices on 2026-07-30 (Luna \$1.00 → \$0.20 per 1M tokens, 5.0×; Terra \$2.50 → \$2.00, 1.2×; grade B, observed), and AWS cut P5 H100 on-demand from more than \$7.50 to \$3.90/GPU-hr in June 2025 (an advertised up-to-45% cut, grade B): the end of the 2023–24 scarcity premium. Over the same era, volumes exploded: ChatGPT WAU 100M → 900M, Anthropic revenue \$1.0B → \$14.0B, OpenRouter 100T aggregate tokens in a single year (November 2024 to November 2025; single-router subset, grade B), and ChatGPT daily prompts 1.0B → 2.5B (December 2024 to July 2025).

A naive arithmetic check shows why the Jevons debate exists: treating the quality-adjusted price fall (5–10×/yr, i.e. $d\ln p = -1.61$ to -2.30/yr) and Anthropic revenue growth (10.6×/yr,

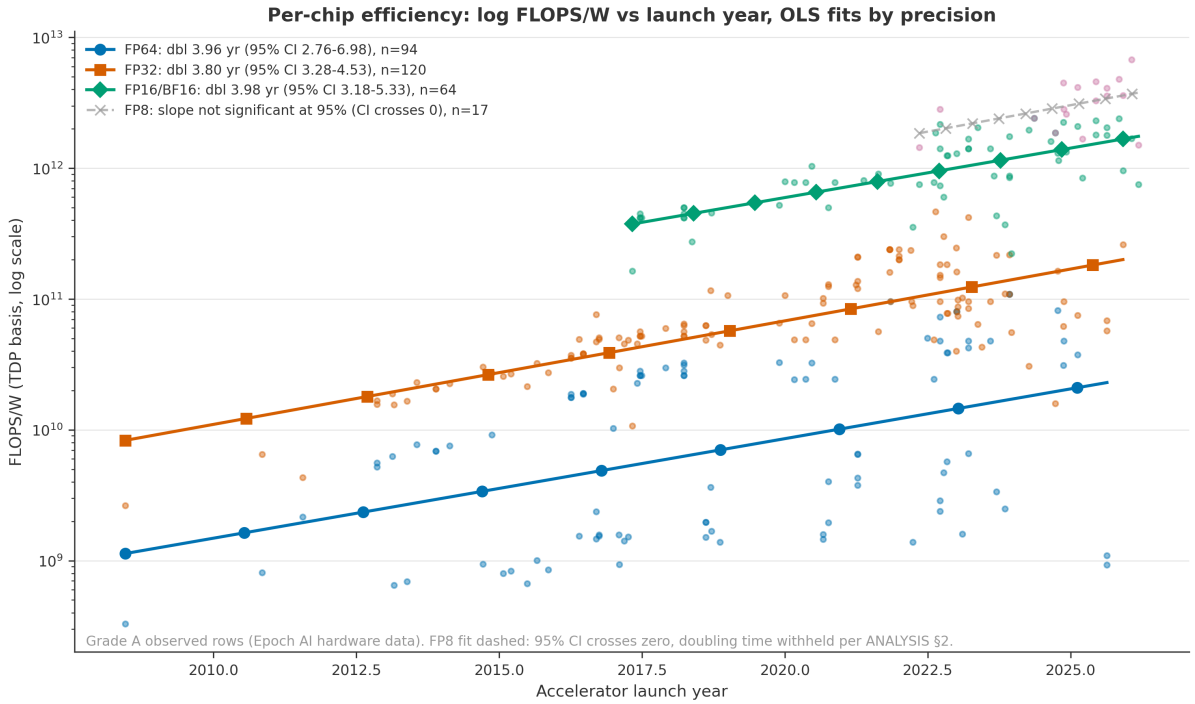


Figure 2: Per-chip efficiency: log FLOPS/W (TDP basis) vs. accelerator launch year, with OLS fits of ln(value) on launch year per precision (Epoch AI machine-learning-hardware data, all grade A observed rows). FP64 (`hw-flops-per-watt-fp64`, $n = 94$, 2008-06-16 to 2025-08-22): doubling 3.96 yr (95% CI 2.76–6.98); FP32 (`hw-flops-per-watt-fp32`, $n = 120$, 2008-06-16 to 2025-12-02): 3.80 yr (3.28–4.53); FP16/BF16 (`hw-flops-per-watt-fp16bf16`, $n = 64$, 2017-05-01 to 2026-03-11): 3.98 yr (3.18–5.33); FP8 (`hw-flops-per-watt-fp8`, $n = 17$, 2022-05-10 to 2026-03-11): slope 0.186/yr with 95% CI (−0.020, 0.392) crossing zero: not significant at 95%, drawn dashed gray with the doubling time withheld per the analysis. Estimates: `eff_fp64_watt_doubling_time`, `eff_fp32_watt_doubling_time`, `eff_fp16bf16_watt_doubling_time`, `eff_fp8_watt_doubling_time`. Caveats: per-chip TDP (vendor thermal design power), not measured sustained draw; precisions never mixed; fit slopes re-verified in-script against `key_estimates.csv` before plotting.

$d \ln E = +2.36$/yr) as market-wide price and expenditure gives $1 - |\varepsilon| = d \ln E / d \ln p$, i.e. $|\varepsilon| \approx 2.47$ (5×/yr) to 2.03 (10×/yr): backfire territory. This is shown in Figure 5 as the gray band, *context only*: it mixes one vendor's revenue with a market-wide price index, ignores quality change already embedded in the price series, and expenditure is not quantity. It is presented to explain the debate, not as an elasticity estimate; the only causal estimate is Musolff's experiment.

Verdict. We do *not* assume Jevons wins. On the price-elasticity evidence, the weights lean to $|\varepsilon| < 1$ (efficiency gains shrink required capacity), with the residual oversupply risk coming from adoption-driven demand undershooting the pipeline, not from backfire. Elasticities are regime- and market-specific; a capability discontinuity (a new killer workload) could shift demand outward faster than efficiency shrinks footprints: the mechanism behind the bull scenario in Section 6.4. Critically for the stranding question: if demand saturates while efficiency keeps improving at ∼20%/yr, the $|\varepsilon| < 1$ regime is exactly the regime that strands capacity: every 1% efficiency gain removes ∼0.5% of the required footprint.

Table 4: Efficiency-curve OLS fits: ln(value) on launch year.

Series	Doubling (yr)	95% CI	Annual (%)	95% CI	n, R^2
FP64 FLOPS/W	3.96	2.76–6.98	19.1	10.4–28.5	94, 0.186
FP32 FLOPS/W	3.80	3.28–4.53	20.0	16.6–23.5	120, 0.566
FP16/BF16 FLOPS/W	3.98	3.18–5.33	19.0	13.9–24.4	64, 0.502
FP8 FLOPS/W	n.s. (slope 0.186/yr)		20.5	−1.9–48.0	17, 0.199
FP64 FLOPS/$	n.s. (slope 0.068/yr)		7.0	−1.0–15.6	53, 0.057
FP32 FLOPS/$	4.74	3.14–9.72	15.7	7.4–24.7	54, 0.228
FP16/BF16 FLOPS/$	3.51	2.11–10.33	21.8	6.9–38.8	20, 0.360
FP8 FLOPS/$	n.s. (slope 0.290/yr)		33.6	−7.2–92.4	9, 0.335

n.s. = slope not significant at 95% (CI crosses zero); headline doubling time withheld. Slopes: FP64/W 0.1752, FP32/W 0.1823, FP16BF16/W 0.1742, FP8/W 0.1862, FP64/$ 0.0679, FP32/$ 0.1462, FP16BF16/$ 0.1975, FP8/$ 0.2899 per year.

Notes: "Annual" $= e^{\text{slope}} - 1$. Doubling time withheld when the slope is not significant at 95%. FLOPS/$ is nominal (launch list prices, no deflator); FLOPS/W is on the TDP basis. All rows grade A, observed.
Source: Epoch AI machine-learning-hardware data; OLS fits by the author.

5.4 Staged supply-demand balances: 2030 under three scenarios

Figure 4 stages the 2030 US pipeline by realization factor; Table 3 gives the scenarios with stated assumptions and falsification conditions. The 2024 operating base is 39.5 GW (Electric Power Research Institute 2026) (catalog-transcribed). Under-construction stages are taken at 1.0 (already financed and digging); announced-advanced and announced-early stages are haircut by published factors: Goldman Sachs high 0.72/0.60 (Goldman Sachs 2026), mid 0.60/0.50, stress ERCOT 0.498 / Carbon Direct PJM 0.24, and JLL 0.95 pre-commitment.

The staged 2030 supply range is **87–152 GW**: stress-realization 86.9 GW (ERCOT 0.498 advanced / PJM 0.24 early), mid path 108.1 GW (GS mid 0.60/0.50), high path 119.3 GW (GS high 0.72/0.60), and max-pipeline variants to 151.5 GW (max pipeline, high factors). The implied 2024–2030 CAGR of the mid-staged path (39.5 → 108.1 GW) is 18.3%/yr nominal capacity.

Balance reading. *Bull (132 GW*: 22.3%/yr demand growth): staged supply 87–152 GW falls short by up to ∼45 GW at the low end (132 − 87); only the max-pipeline/high-realization corner (151.5 GW) covers it. Persistent tightness: vacancy stays near 1%, lease rates keep rising. *Base (96 GW*: 16.0%/yr): roughly balanced; mid-staged 108 GW covers it with ∼12 GW headroom (108 − 96); the stress-staged path (86.9 GW) leaves a ∼9 GW shortfall (87 − 96). Near-term tight, medium-term balanced. *Bear (56 GW*: 6.0%/yr): structural oversupply of 31–96 GW: 87 − 56 = 31 at the low end, 152 − 56 = 96 at the high end. This is the scenario in which efficiency gains ($|\varepsilon| < 1$) plus adoption saturation strand 30–95 GW of pipeline.

Cross-check. Goldman Sachs expects ∼95 GW installed by end-2027 (∼2× end-2025, at 70% utilization; `gs_us_dc_installed_capacity`, grade B) (Goldman Sachs 2026); on the base path, EPRI implies ∼68 GW by 2027 (linear interpolation of its stated linear 2024–2030 pathway: 39.5 GW in 2024 to 96 GW in 2030); GS is the more aggressive forecaster, its end-2027 figure already near EPRI's 2030 base level. GS's power-demand path of 31 → 41 → 66 GW (2025 → 2026 → 2027, 45.9%/yr) is consistent with a bull-leaning near term.

5.5 Early-warning dashboard

Table 4 and Figure 6 give six current readings, each with date, grade, and a TIGHTNESS / SOFTNESS / MIXED classification. The net read: *the near-term physical market is still tight,*

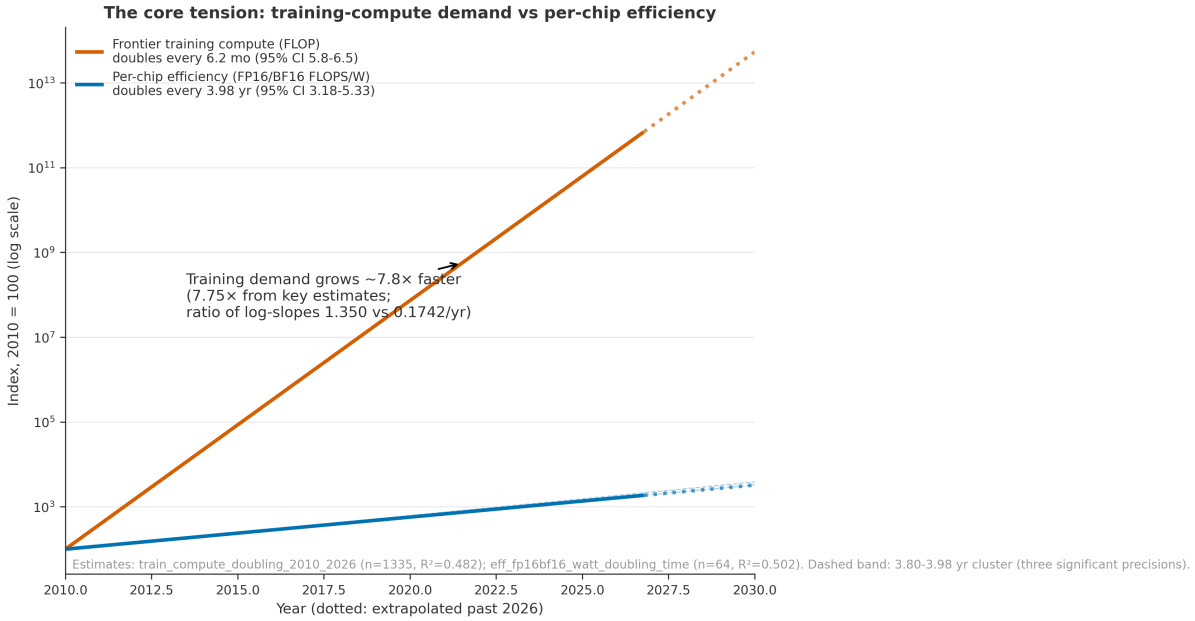


Figure 3: The core tension: frontier training-compute demand vs. per-chip efficiency on one log-time index (2010 = 100). Training compute (`epoch-training-compute`, DL-era 2010-01-01 to 2026-09-09, $n = 1{,}335$, grade A): log-slope 1.350/yr, doubling every 6.2 months (95% CI 5.8–6.5; `train_compute_doubling_2010_2026`, $R^2 = 0.482$). Per-chip efficiency (FP16/BF16 FLOPS/W, the AI-relevant precision): log-slope 0.174/yr, doubling every 3.98 yr (95% CI 3.18–5.33; `eff_fp16bf16_watt_doubling_time`, $n = 64$, $R^2 = 0.502$). Training demand grows ∼7.8× faster (7.75× from the key estimates; ratio of log-slopes). Dashed band: the 3.80–3.98 yr doubling-time cluster across the three precisions with significant fits. Dotted segments past 2026 are extrapolations, not observations.

but every price signal at the margin points to softening.

On the demand side, the quantity signals are unambiguously strong: vacancy fell from 3.7% to 1.6% to **1.4%** in 2025 (a historic low, despite +36% capacity growth), net absorption rose from 1,809.5 to **2,497.6** MW in 2025 (+38% YoY; 2024 was +450% vs 2020; Northern Virginia alone absorbed 1,102 MW of the 2025 total), and JLL's preleased share of the pipeline rose from 73% (2025-H1) to **92%** (2025-YE), with 9 GW delivered in 2025 and 35 GW under construction. On the price side, everything at the margin softens: asking lease-rate growth decelerated from +18.6% to +7.0% to +12.6% to **+6.5%**; the level still rises ($194.95/kW/mo) but growth is slowing (MIXED); AWS cut P5 H100 on-demand by an advertised up-to-45% (more than $7.50 → **$3.90**/GPU-hr, June 2025), ending the 2023–24 hyperscaler scarcity premium on list pricing (SOFTNESS, demand-side); prior-gen GPU cheapest-listed prices fell 19.0% (A100) and 33.2% (L40S) in 30 days to floors of $0.53/hr and $0.67/hr on Vast.ai (2026-09; grade C, labeled context, not a headline); and H100 marketplace prices dispersed 21× ($0.72–$15.14/hr in one 24h window, 141k+ observations) against a hyperscaler list of ∼$12.29/GPU-hr (SOFTNESS/fragmentation in the marketplace tier). Forward-looking: Microsoft cancelled or deferred ∼2 GW of leases in the US and Europe over the prior six months (2025-03-26 event), attributed to oversupply vs. updated forecasts plus reduced incremental OpenAI training demand (SOFTNESS, demand-side; partially offset by Google/Meta backfilling some released capacity per TD Cowen); Bernstein estimates 35–40% of announced global capacity is at risk of delay or cancellation (2026-07 event), while flagging cost/MW up 20–25% since 2023 and transformer lead times of 80–100 weeks (SUPPLY-SIDE constraint: power/equipment gating, not demand withdrawal); and CBRE NA under construction fell from 6,350.1 to **5,994.4** MW (−5.6%) in 2025, the first decline since 2020, attributed in the source notes to permitting/zoning/power delays, not demand withdrawal.

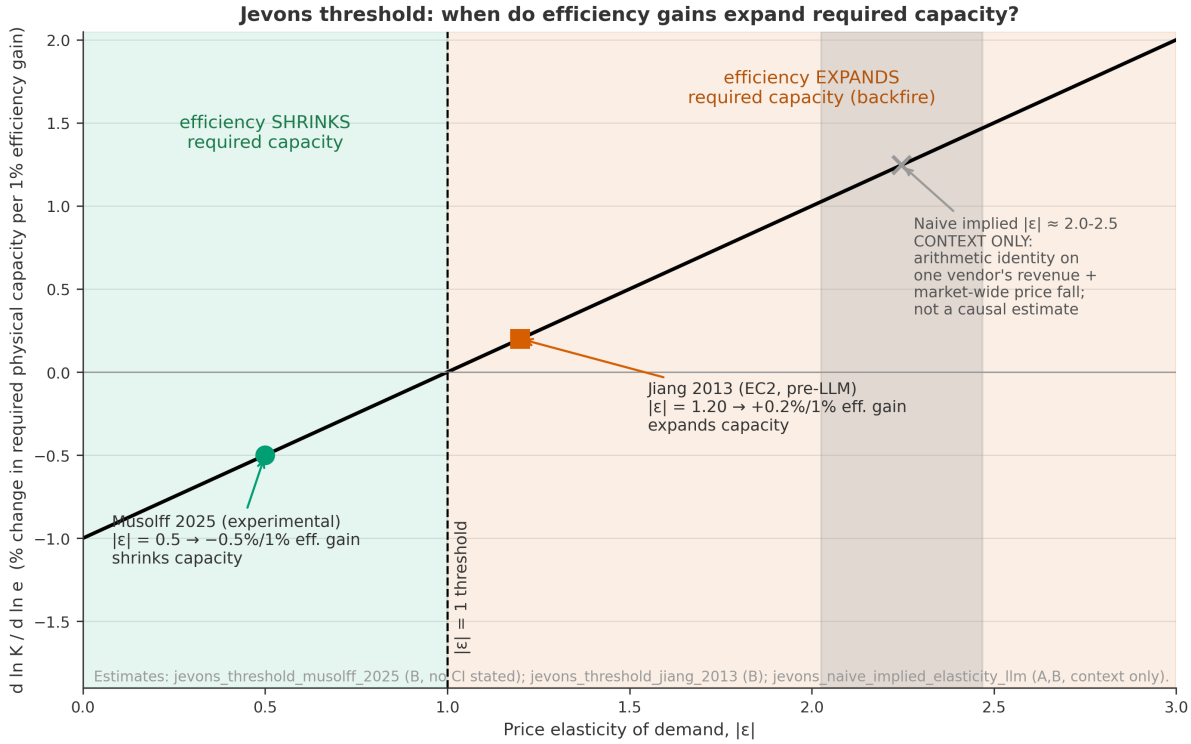


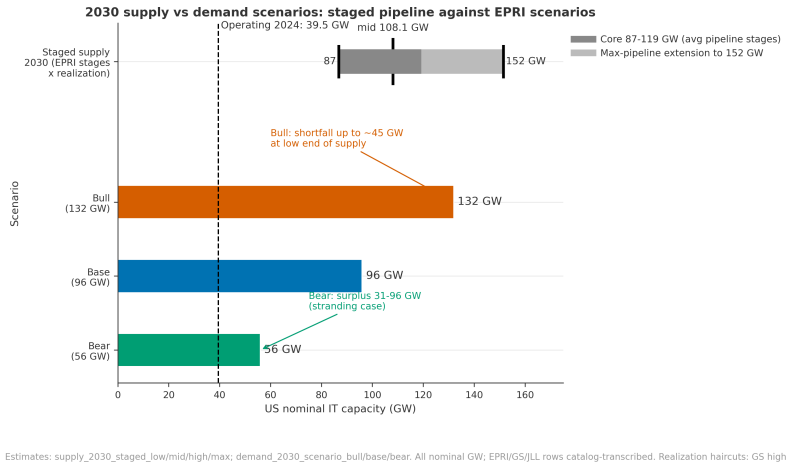
Figure 4: Jevons threshold diagram: $d\ln K/d\ln e = |\varepsilon| - 1$ vs. price elasticity of demand $|\varepsilon|$, where K is required physical capacity and e is efficiency (a 1% efficiency gain changes required capacity by $(|\varepsilon| - 1)\%$). Green region ($|\varepsilon| < 1$): efficiency shrinks required capacity; orange region ($|\varepsilon| > 1$): backfire. Marked: Musolff 2025 experimental estimate $|\varepsilon| = 0.5 \to -0.5\%/1\%$ gain (`jevons_threshold_musolff_2025`, grade B; no numeric CI stated; cloud spot compute, not genAI inference); Jiang 2013 EC2 $|\varepsilon| = 1.20 \to +0.2\%/1\%$ gain (`jevons_threshold_jiang_2013`, grade B, pre-LLM era); $|\varepsilon| = 1$ threshold (dashed). Gray band: the naive arithmetic implied $|\varepsilon| \approx 2.0$–$2.5$ (`jevons_naive_implied_elasticity_llm` $= 2.247$, CI 2.026–2.468), shown as context only: it mixes one vendor's revenue growth with a market-wide price fall and is not a causal estimate.

The softness is concentrated in prior-gen silicon, the forward pipeline, and marketplace fragmentation, exactly where the analogue templates say the first cracks show.

5.6 Historical analogues and indicator ordering

Each analogue formalizes which indicator turned first, and the measured lags differ by analogue: telecom equity and capex peaks preceded Global Crossing's filing by about 1–2 years while the unit-price peak led by about nine; merchant power's 2001 price spike preceded Calpine's 2005 filing by about 4.9 years; LNG utilization collapsed over 2007–2012 with no comparable financial-distress wave; railways absorbed the price collapse without a glut-driven distress wave. Capex is a lagging indicator; contracted revenue is coincident at best. Figure 7 shows the three quantitative templates.

Telecom / fiber, 1990–2007. Ordering: unit prices and equity → capex → distress. Retained revenue per interstate minute peaked in 1993 ($0.117) and fell to $0.046 by 2006 (−61%; grade A, nominal) (Federal Communications Commission 2008), the NASDAQ telecom index peaked at 1,230.06 on 2000-03-10 and fell 62% by year-end 2000 (grade B), and a London–New York circuit fell from $22,000 (start 2001) to $5,000 (Aug 2002), with a forward quote near



Estimates: supply_2030_staged_low/mid/high/max; demand_2030_scenario_bull/base/bear. All nominal GW; EPRI/GS/JLL rows catalog-transcribed. Realization haircuts: GS high 0.72 adv / 0.60 early; GS mid 0.60 / 0.50; stress ERCOT 0.498 / PJM 0.24 (UC taken at 1.0).

Figure 5: 2030 US nominal IT capacity: staged supply vs. EPRI demand scenarios (GW). Demand bars: bear 56, base 96, bull 132 GW (`demand_2030_scenario_bear/base/bull`, `epri_us_nominal_it_capacity_scenario_low/medium/high` 2030-01-01, grade A catalog-transcribed). Supply bar: staged 2030 range 87–152 GW, core 87–119 GW (hatched; avg-pipeline stages × realization factors), mid path 108.1 GW tick, light-gray extension to 152 GW (max-pipeline variants); estimates `supply_2030_staged_low/mid/high/max` (86.9/108.1/119.3/151.5 GW). Dashed line: operating 39.5 GW in 2024 (`epri_us_nominal_it_capacity_operating`, grade A catalog-transcribed). Realization haircuts as published: GS high 0.72 advanced / 0.60 early, GS mid 0.60 / 0.50, stress ERCOT 0.498 / PJM 0.24; under-construction taken at 1.0. Readings: bull leaves a shortfall up to ∼45 GW at the low end of supply; base roughly balanced (∼12 GW headroom at mid); bear implies 31–96 GW structural surplus (the stranding case). Caveats: all EPRI/GS/JLL capacity rows are catalog-transcribed ("transcribed from catalog, verify from primary"); all values nominal GW.

$2,000, a −77% collapse from start-2001 to August 2002 (grade B, nominal). Broadband-provider capex peaked in 2000 ($118.1bn), troughed in 2003 ($57.0bn), and never regained the 2000 peak through 2017 (grade B, nominal) (USTelecom 2018), while industry after-tax profit went from +$21.6bn (1996) to −$27.9bn (2000) (grade B). The circular-demand tell: CLEC revenue grew from $3.35bn (1998) to $16.93bn (2005, grade A) while interexchange revenue fell from $87.3bn (2000) to $46.9bn (2005, grade A); entrants grew on the incumbents' traffic even as the core business halved. Leads to the first major distress filing (Global Crossing Ch. 11, 2002-01-28): unit-price peak (1993) ≈ 8.6 yrs; equity peak (2000-03-10) ≈ 1.9 yrs; capex peak (2000) ≈ 1.6 yrs (annual peaks dated at mid-year; dated events exact).

Merchant power, 1999–2005. Ordering: (false) price signal → construction pipeline → utilization → distress. The California wholesale price spike (>300/MWh in early 2001; grade B, lower bound) financed a pipeline of 54,120 MW of planned 2001 US additions (grades B/C); Calpine's fleet utilization broke to ∼45% in 2005 (grade C, labeled) and Calpine filed Ch. 11 in December 2005 (grade B): price spike to distress ≈ 4.9 yrs. (Enron, Dec-2001, grade C; PG&E, Apr-2001, is flagged in the source notes as a market-design failure, not glut evidence.)

LNG import terminals, 2004–2012. Ordering: utilization collapse was the earliest and clearest warning. All-CONUS import cargo counts fell from 274 (2007) to 131 (2008) to 58.3 (2012), a 78.7% collapse (grade A); Cove Point went from 78 (2005) to 2 (2012); Lake Charles long-term from 139.6 Bcf (2007) to 2.5 Bcf (2012), while the EIA's January 2007 forecast still showed 770 Bcf (2007) and 1,080 Bcf (2008) (grade A): expectations rose as utilization collapsed. New terminals took first cargos in 2008–2011 into a vanished market; no financial-distress wave

Table 5: 2030 US nominal IT capacity (GW): demand scenarios vs. staged supply.

	2030 demand (GW)	Demand CAGR	2030 supply (GW)	Balance vs. mid 108 GW
Bull	132	22.3%/yr	87–152	Shortfall up to ∼45 GW at low end
Base	96	16.0%/yr	87–152	∼12 GW headroom at mid
Bear	56	6.0%/yr	87–152	Surplus 31–96 GW (stranding case)

Assumptions (all scenarios): AI-relevant efficiency improves ∼20%/yr (mean of the FP16/BF16 and FP8 FLOPS/W fits, Section 6.2); all figures nominal GW. **Bull** (132 GW): Jevons backfire ($|\varepsilon| > 1$) or a capability-driven adoption discontinuity; efficiency expands demand faster than it shrinks footprints. *Falsified if* 2027–28 installed-capacity growth stalls below 22%/yr while frontier GPU list prices stay flat or fall. **Base** (96 GW): $|\varepsilon|$ ∼0.5–1 (Musolff-like); efficiency modestly shrinks the physical footprint; growth is adoption-driven. *Falsified if* net absorption turns negative or vacancy rises above ∼4% with lease rates declining YoY. **Bear** (56 GW): $|\varepsilon| < 1$ and demand saturates; efficiency outruns adoption; AI workloads follow the classic enterprise-IT S-curve. *Confirmed if* vacancy exceeds 8% sustained with frontier list-price cuts and under-construction cancellations.

Notes: Demand: EPRI 2026 scenarios (grade A, catalog-transcribed). Supply: staged by published realization factors (Section 5).
Source: Demand scenarios (Electric Power Research Institute 2026); supply staging by the author (Section 5).

followed (absorption via conversion to export).

Railways, 1860–1920. Ordering: rates and financing lead; volumes are a lagging confirmation: the absorption case. Freight revenue per ton-mile fell from 1.236¢ (1882) to 0.726¢ (1899), −41.3% (grade A-, nominal; no pre-1947 CPI, so within-series ratios only), while throughput kept growing: all-rail ton-miles 39.3 → 252.9 bn (1882 → 1911, ∼6.4×, grade A-) and 13 majors 2.16 → 17.83 bn ton-miles (1865 → 1885, ∼8.3×, grade B). Mileage growth broke only after 1890 (163,581 → 193,321 → 240,414 → 248,700, decadal 1890–1920). Distress was macro-driven (121 Panic-of-1873 railroad failures, grade C narrative), not a capacity-glut wave. No stranding: the capacity was absorbed.

Mapping the AI buildout onto each template (as of 2026-09-25). The buildout is *pre-distress in all templates*. The closest matches: (1) telecom 1998–99: margin price softness (marketplace / prior-gen) while the build accelerates and incumbents report strength, with a circular-demand echo (neoclouds / CLEC-like entrants growing on hyperscaler capacity); (2) merchant power 2000–01: price signals financing a pipeline before utilization is tested, with power and transformers (not capital) as the gating factor, exactly as in 2001. The key difference from telecom: demand revenue is still accelerating (OpenAI ∼216%/yr, Anthropic ∼10×/yr, absorption +38% YoY); interexchange revenue was already collapsing in 2000. LNG's lesson is the sharpest: utilization was the earliest warning and it is currently the one datum we cannot observe: the first observable crack to watch for is sustained declines in frontier-GPU list prices or preleased shares (the utilization proxies whose measured lead over distress ranges from about 1–2 years (telecom equity/capex) to about 9 years (telecom unit prices) to about 4.9 years (merchant power)). The railway template is the bull case: rates can collapse 41% and volumes still absorb everything; it requires demand (ton-miles then, tokens/tasks now) to keep growing ∼6× through the price decline.

Table 6: Early-warning dashboard (as of 2026-09-25).

Signal	Reading	Date	Grade	Class
CBRE NA primary vacancy	3.7% → 1.6% → **1.4%**	2025	B	TIGHTNESS
CBRE NA lease-rate YoY growth	+18.6 → +7.0 → +12.6 → **+6.5%**	2025	B	MIXED
JLL NA preleased share of pipeline	73% (H1) → **92%** (YE)	2025	B	TIGHTNESS
CBRE NA net absorption	1,809.5 → **2,497.6** MW	2025	B	TIGHTNESS
AWS P5 H100 on-demand list	$7.50 → **$3.90**/GPU-hr (−45%)	2025-06	B	SOFTNESS
Prior-gen GPU cheapest 30-day deltas	A100 **−19.0%**, L40S **−33.2%**	2026-09	C	SOFTNESS
H100 marketplace 24h price dispersion	**21**× ($0.72–$15.14/hr)	2026-05-13	B	SOFTNESS
Microsoft lease cancellations	**∼2 GW** (US+Europe)	2025-03-26	B*	SOFTNESS
Bernstein: announced capacity at risk	**35–40%**	2026-07	B*	SUPPLY-SIDE
CBRE NA under construction	6,350.1 → **5,994.4** MW (−5.6%)	2025-YE	B	SUPPLY-SIDE

Vacancy is a historic low despite +36% capacity growth; lease growth decelerating (4th straight annual increase, now +6.5%; $194.95/kW/mo asking, 250–500 kW N+1/Tier III). Absorption +38% YoY (Northern Virginia 1,102 MW of the 2025 total; 2024 was +450% vs 2020). Hyperscaler list ∼$12.29/GPU-hr sits far above the marketplace (AIMC, 141k+ obs). Bernstein flags cost/MW +20–25% since 2023 and transformer lead times of 80–100 weeks; the 2025 UC decline is the first since 2020 and is attributed to permitting/zoning/power delays per the source notes, not demand withdrawal. Microsoft's released capacity was partially backfilled by Google/Meta (TD Cowen). The documented gap is that no public AI-fleet GPU utilization series exists: Uptime rows measure the share of operators *collecting* utilization, not utilization itself.

Notes: B* = event rows from `events_pricing.csv` (13 events: 9 demand-driven, 4 supply-driven); grade of the underlying press source.
Source: CBRE, JLL, AWS, GPU marketplace data, Bernstein, TD Cowen, and press event rows; full notes per row in `data/master_pricing_demand.csv` and `data/events_pricing.csv`.

6 Robustness and limitations

Nominal rates. No deflator was retrieved for capex, efficiency, or pricing rows (the Elkassabgi-Data FRED/BEA probe failed with a provider HTTP error in phase 1), so every growth rate in this paper is nominal. Cumulative US inflation over the window is substantial and flatters the nominal CAGRs; the ordering of results is unaffected but absolute levels are not real-terms. The one exception is source-provided real dollars (Epoch training cost, 2023 USD), which are never mixed with nominal series.

The JLL-vs-EPRI under-construction inconsistency. Figure 8 shows the unresolved inconsistency honestly. JLL reports 66 GW under construction in North America (H1 2026; `jll_na_under_construction`, grade B, catalog-transcribed) vs EPRI's US under-construction avg 12.9 GW / max 17.2 GW (`epri_us_capacity_pipeline_under_construction_avg/max`, grade A, catalog-transcribed): a ∼4–5× gap that scope (NA vs US) cannot explain; one or both definitions differ materially. The staged 2030 supply build (Figure 4) uses EPRI's internally consistent stage breakdown and carries the JLL figure as a sensitivity input: if 66 GW of UC realizes at ∼1.0, 2030 supply is ≈ 39.5 + 66 + realized-announced ≈ 140 GW (the announced stages are included in that arithmetic, at the low-stress realization factors), a material upside to the supply range that would deepen the bear-case surplus and narrow the bull-case shortfall. Both values are transcribed from catalog; verify from the primary reports before use.

The utilization gap. No public AI-fleet GPU utilization series exists. The Uptime rows measure the share of operators *collecting* utilization, not utilization itself; preleased share (92%) and vacancy (1.4%) are the available proxies. Per the LNG template, utilization was the single

earliest observable warning; its unobservability is this study's binding constraint, and it is the reason the dashboard leans on price proxies.

Elasticity evidence. Musolff's −0.5 carries no numeric CI in the scout pass and is cloud spot compute, not genAI inference; Jiang's 1.20 is 2013 pre-LLM. No causal elasticity estimate exists for LLM inference specifically: the paper's inference-market conclusions rest on the observed adoption-vs-price decomposition (flat buyer price per completed task alongside exploding volumes), not on an elasticity estimate we do not have.

Transcription and dating. All EPRI/GS/JLL/Synergy/Carbon Direct capacity rows are catalog-transcribed (flagged "not independently re-verified from raw/" in every row's notes); that caveat carries into the staged supply build. NVIDIA FY2023 and FY2024 rows share the date 2024-01-28 (distinguished by FY label in notes); intermediate annual growth rates between FY2022 and FY2024 should be read from FY labels, not dates; the endpoints used here (FY2016/FY2026) are unaffected. The llm-frontier weekly medians are noisy (26–180 model-night observations/week) and its model slugs do not correspond to known commercial models; used only for event dating, not levels. MLQ 30-day deltas and hemgo-lexsi daily medians are grade C and labeled as context, not headlines.

Scope of the capex perimeter. The hyperscaler total covers all cash capex, not capex attributed to AI: hyperscalers do not disclose an AI split, and any AI share would be an assumption. Tencent is excluded (not SEC-registered; HKEX annuals not extracted). Financing events (CoreWeave, Crusoe, neoclouds) are ingested only with exact figures and usable dates; announcements were never upgraded (the full exclusion log is in the capex build documentation).

7 Discussion: causal mechanisms and implications

7.1 What mechanism could strand capacity?

Three mechanisms are consistent with the evidence, and they point in different directions. *(i) Jevons backfire* ($|\varepsilon| > 1$): ruled out on the only causal estimate, but the naive arithmetic ($|\varepsilon| \approx 2.0$–$2.5$) explains why the intuition persists: it conflates adoption-driven volume growth with price-driven rebound. *(ii) The quality-expansion channel*: efficiency and capability gains enable new use cases, shifting the demand curve outward rather than moving along it. This is the mechanism behind both the bull scenario and the observed LLM growth (flat buyer price per task, exploding volumes); and it is reversible: if capability gains slow, the outward shifts stop while efficiency keeps shrinking the required footprint. *(iii) The pipeline-overshoot channel*: announced capacity is financed against forecasts (EPRI bull: 132 GW) while realization factors (ERCOT 0.498, PJM 0.24, Bernstein 35–40% at risk) show that a third or more of announced capacity never materializes. The bear case is mechanism (ii) stalling plus (iii) delivering: demand saturates on the enterprise S-curve while staged supply arrives at 87–152 GW. The best-supported stranding mechanism is therefore *efficiency plus adoption undershoot*, not Jevons backfire, the same combination the $|\varepsilon| < 1$ regime predicts.

7.2 Implications for forecasting

Forecasts are the mechanism of overbuild in every analogue (EIA's 1,080 Bcf, Deutsche Banc's 2001 reserve margins, Global Crossing's enterprise-bandwidth pitch). Three disciplines follow. First, never take announced capacity at face value: stage it by published realization factors, as in Section 6.4, and carry definitional inconsistencies (Figure 8) as explicit sensitivities. Second, attach falsification conditions to every demand path (Table 3) and read them off a dashboard

whose indicator ordering is measured per analogue (telecom ∼1–2 yrs for equity/capex, ∼9 yrs for unit prices; merchant power ∼4.9 yrs), not a uniform 1–3-year rule. Third, keep the nominal/real distinction: without a deflator, the 31.6%/yr capex CAGR is flattered by inflation, and any real-terms claim built on it is an assumption, not a finding.

7.3 Implications for industrial strategy

The gating factor in 2026 is power and equipment, not capital: transformer lead times of 80–100 weeks, cost/MW up 20–25% since 2023, and permitting/zoning delays explain the first UC decline since 2020. This mirrors merchant power in 2001, where the binding constraint shifted from financing to interconnection and equipment. Strategy that treats the bottleneck as money will misfire; the scarce inputs are grid access, transformers, and entitled land with power. The circular-demand tell from telecom (entrants growing on incumbents' traffic) has its AI echo in neoclouds and capacity resellers: the share of "demand" that is carrier-to-carrier rather than end-user is the series to watch, and it is not currently published.

7.4 What the results imply for underwriting the buildout

For readers assessing the buildout's funding, the paper's results point toward underwriting *demand realization*, not capacity announcements: contract quality (preleased share, counterparty credit), utilization proxies (frontier-GPU list prices, lease-rate momentum), and demand-revenue series (OpenAI and Anthropic revenue, absorption) rather than headline capex. The analogue lead times: 1.9 yrs (telecom) to 4.9 yrs (merchant power) from price-signal turn to distress, setting the monitoring horizon: a dashboard read quarterly, not a forecast believed annually.

8 Conclusion: is the boom funded by real demand, and what would prove it wrong?

Is the boom funded by real demand? In the near term, yes: realized AI demand revenue is accelerating (OpenAI 216%/yr, Anthropic ∼10×/yr, ChatGPT 100M → 900M WAU), net absorption rose 38% YoY to 2,497.6 MW, vacancy sits at a historic-low 1.4%, and 92% of the pipeline is preleased. The physical market is genuinely tight today, and the demand-revenue series are observed, not forecast.

But is the *pipeline* justified? The pipeline is sized for the base-to-bull demand path. Staged 2030 supply of 87–152 GW roughly covers base demand (96 GW, ∼12 GW headroom at mid), falls short under bull conditions (132 GW, shortfall up to ∼45 GW), and is structurally oversupplied under bear conditions (56 GW, surplus 31–96 GW). The margin price signals sit exactly where the telecom 1998–99 and merchant-power 2000–01 templates say the first cracks show: the 45% hyperscaler GPU price cut, 21× marketplace dispersion, prior-gen price collapse, decelerating lease growth, and 35–40% of announced capacity at risk.

Does efficiency strand capacity? On the best causal evidence ($|\varepsilon| = 0.5$), a 1% efficiency gain *shrinks* required physical capacity by ∼0.5%; efficiency is not the enemy of utilization in the price channel. The stranding risk is the interaction the paper formalizes: if adoption saturates while efficiency keeps improving at ∼20%/yr, the $|\varepsilon| < 1$ regime steadily removes required footprint from under a fixed pipeline. Jevons backfire is the wrong mechanism to fear; efficiency outrunning adoption is the right one.

What would prove this wrong? The study is falsifiable on its own dashboard. The "balanced at base, surplus under bear" reading fails if: (a) sustained declines in frontier-GPU list prices (the closest public proxy for the unobservable utilization series) coincide with preleased shares falling below ∼70% and vacancy rising past 4%: the utilization-proxy turn whose measured lead over distress ranges from about 1–2 years (telecom equity/capex) to about 9 years (telecom unit prices) to about 4.9 years (merchant power); (b) demand-revenue growth decelerates below the base path (16%/yr to 2030) while staged supply delivers at the mid path or above; or (c) a capability discontinuity shifts demand outward faster than efficiency shrinks footprints, validating the bull case. Watch the prices and the utilization proxies. Capex will tell you last.

References

J. S. Armstrong, editor. *Principles of Forecasting: A Handbook for Researchers and Practitioners*. Kluwer Academic Publishers, 2001.

C. Buffington, L. Foster, et al. Measuring the adoption of AI: Evidence from the business trends and outlook survey. Technical Report CES 24-16, U.S. Census Bureau, 2024.

Carbon Direct. AI meets the grid: Interconnection queue analysis in PJM and ERCOT. Carbon Direct, May 14, 2026.

E. Couper, J. Hejkal, and A. Wolman. Boom and bust in telecommunications. *Federal Reserve Bank of Richmond Economic Quarterly*, 89:1–24, 2003. Fall 2003.

Deutsche Banc Alex. Brown. US utilities. June 5, 2001.

Electric Power Research Institute. Powering intelligence 2026: Updated scenarios of U.S. data center electricity use and power strategies. Technical report, Electric Power Research Institute, 2026. Successor to the 2024 edition, product 3002028905.

Electric Reliability Council of Texas. Long-term load forecast update 2025–2031 and methodology changes. Technical report, 2025. April 7, 2025; ERCOT Board of Directors meeting, Dec 9, 2025 (large-load queue status).

Epoch AI. Data on machine learning hardware. https://epoch.ai/data/machine-learning-hardware (online resource); accessed 25 Sep 2026, 2024.

Epoch AI. Data on AI models. https://epoch.ai/data/ai-models (online resource); accessed 25 Sep 2026, 2025.

Epoch AI. Data on AI chip sales. Published online at https://epoch.ai; accessed 25 Sep 2026. Raw: epoch_ai_chip_sales/, 2026.

Federal Communications Commission. Trends in telephone service. Editions DOC-284932A1, DOC-301823A1; Chart/Table 13.1 and carrier-revenue tables, 2008.

Federal Reserve Bank of Richmond. Seeing double: An AI bubble? *Macro Minute*, October 7, 2025; https://www.richmondfed.org/research/national_economy/macro_minute/2025/seeing_double_an_ai_bubble, 2025.

Goldman Sachs. Commodities team power market report. Goldman Sachs Global Investment Research, May 5; *GS SUSTAIN: Data Center Power Demand: The 6 Ps Driving Growth and Constraints*, Oct 13, 2025, 2026.

R. J. Gordon. *The Rise and Fall of American Growth: The U.S. Standard of Living since the Civil War*. Princeton University Press, 2016.

H. Herring and S. Sorrell, editors. *Energy Efficiency and Sustainable Consumption: The Rebound Effect*. Palgrave Macmillan, 2009.

C. Hogendorn. Excessive(?) entry of national telecom networks, 1990–2001. Working paper, June 13, 2007.

International Energy Agency. Energy and AI. Technical report, International Energy Agency, Paris, 2025. April 2025 (CC BY 4.0).

International Energy Agency. Key questions on energy and AI. Technical report, International Energy Agency, 2026.

W. S. Jevons. *The Coal Question: An Inquiry Concerning the Progress of the Nation, and the Probable Exhaustion of Our Coal-Mines*. Macmillan, London, 1865.

Q. Jiang, Y. C. Lee, and A. Y. Zomaya. Price elasticity of the enterprise computing resource market. Working paper; https://media.qyjohn.net/wp-content/uploads/2015/10/price-elasticity-enterprise.pdf, 2015.

Jones Lang LaSalle. North america data center report – midyear 2026. JLL, Aug 11, 2026.

Menlo Ventures. 2025: The state of generative AI in the enterprise. November, 2025.

L. Musolff, Z. Zhang, J. Hofman, and S. Roy. Reliability and pricing in cloud computing. Working paper, November; https://lmusolff.com/papers/Reliability_and_Pricing_in_Cloud_Computing.pdf, 2025.

B. Nagy, J. D. Farmer, Q. M. Bui, and J. E. Trancik. Statistical basis for predicting technological progress. *PLoS ONE*, 8(2):e52669, 2013.

NBER. Macrohistory database. Series a03003d / a03003e (railroad revenue per ton-mile; revenue ton-miles, 1882–1911); FRED series A0303DUSA259NNBR, 2026.

W. D. Nordhaus. Two centuries of productivity growth in computing. *Journal of Economic History*, 67(1):128–159, 2007.

OpenRouter. 100-trillion-token traffic study. arXiv:2601.10088, 2026.

A. Shehabi, S. J. Smith, A. Hubbard, A. Newkirk, N. Lei, M. A. B. Siddik, B. Holecek, J. Koomey, E. Masanet, and D. Sartor. 2024 united states data center energy usage report. Technical Report LBNL-2001637, Lawrence Berkeley National Laboratory, 2024.

S. J. Smith, A. Hubbard, A. Newkirk, M. Ganeshalingam, B. Holecek, D. Sartor, M. Mills, and A. Shehabi. United states data center energy usage report: 2025 update. Technical report, Lawrence Berkeley National Laboratory, 2025. June 2026.

S. Sorrell. The rebound effect: an assessment of the evidence for economy-wide energy savings from improved energy efficiency. Technical report, UK Energy Research Centre, 2007. October.

Stanford Institute for Human-Centered AI. AI index 2026. Technical report, 2026. Chapter 4 (Economy).

Uptime Institute. 15th annual global data center survey. July 30 ($n > 800$ owners/operators, surveyed Apr–May 2025), 2025.

US Congress Joint Economic Committee. Economic repercussions of the stock market bubble. July 14, 2003.

US Energy Information Administration. Short-term energy outlook supplement. Technical report, US Energy Information Administration, 2007. January: "U.S. LNG Imports — The Next Wave".

US House Financial Services. Market conditions of the U.S. telecom sector, as illustrated by the bankruptcy of Global Crossing, Inc. Written testimony; https://financialservices.house.gov/media/pdf/032102wm.pdf, 2002.

USTelecom. Historical capital expenditures by U.S. broadband providers. Research Brief, October 18, 2018.

T. P. Wright. Factors affecting the cost of airplanes. *Journal of the Aeronautical Sciences*, 3(4): 122–128, 1936.

A Dataset and reproducibility

The analysis dataset comprises six master files (row counts exclude headers):

- data/master_capex.csv: 597 rows, 17 series
- data/master_capacity.csv: 155 rows, 139 series
- data/master_efficiency.csv: 3,742 rows, 264 series
- data/master_pricing_demand.csv: 242 rows, 93 series
- data/events_pricing.csv: 13 dated events
- data/master_analogues.csv: 389 rows

Transformation logs are in data/BUILD_capex.md, data/BUILD_capacity.md, data/BUILD_efficiency.md, data/BUILD_pricing_demand.md, and data/BUILD_analogues.md. The quantitative analysis is generated by the seeded, deterministic results/analyze.py, with headline estimates in results/key_estimates.csv (method and row ranges per estimate) and the full derivation in results/ANALYSIS.md. Figures are produced by the seeded figures/figures.py, which asserts every plotted headline number against key_estimates.csv and raises on mismatch. Every quantitative claim in this paper is indexed in paper/CLAIM_INDEX.md with its estimate_id or source.

Production note. This paper was typeset in LaTeX (11pt, 1-inch margins) from paper/paper.tex; all figures are 300 DPI PNGs produced by the seeded figures/figures.py under the shared institutional chart style; estimation by the seeded results/analyze.py. The PDF is the version of record. Research design, statistical analysis, and drafting were carried out through a quantitative research pipeline with AI assistance; every estimate, figure, and quantitative claim was verified against the source data before publication.

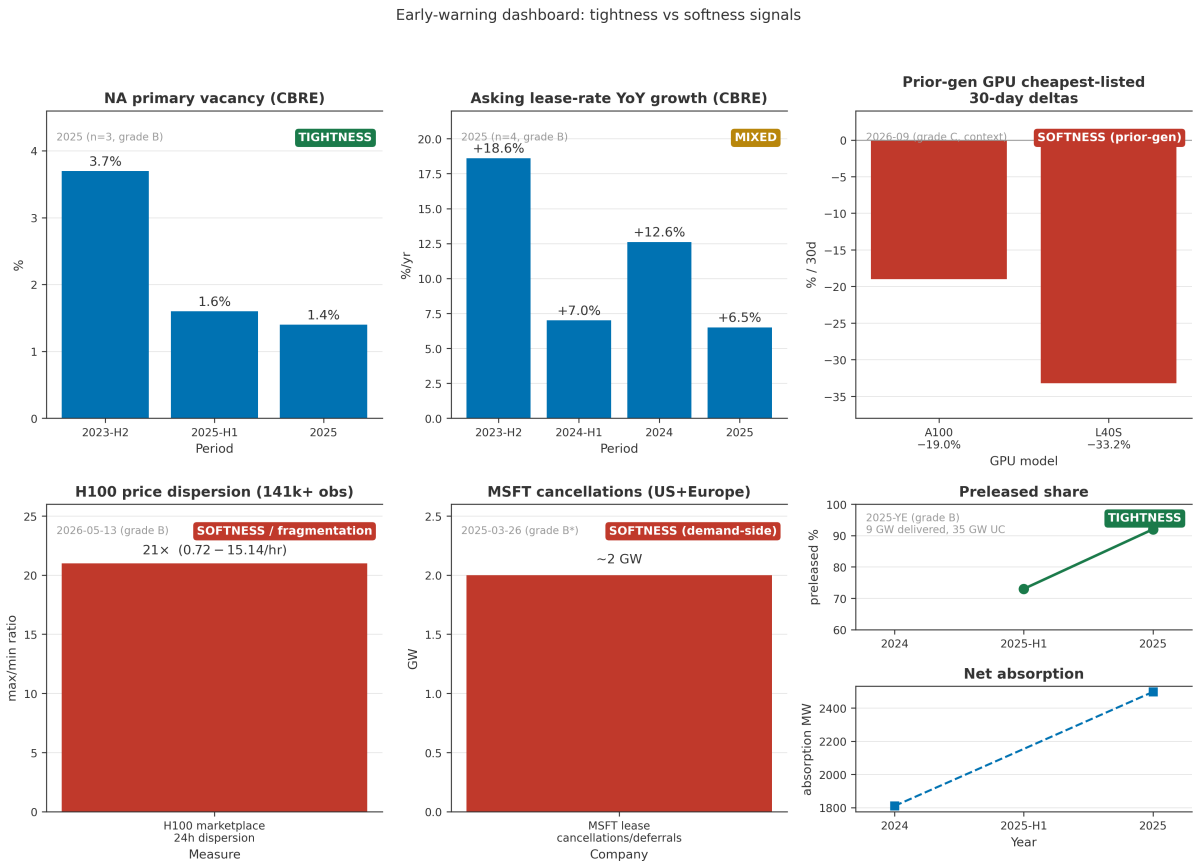


Figure 6: Early-warning dashboard: six current readings, each classified as a tightness or softness signal with date and grade. (a) CBRE NA primary vacancy 3.7% → 1.6% → 1.4% (`vacancy_2025_1_4pct`; `cbre-na-primary-vacancy`, 2023-H2 to 2025, $n = 3$, grade B): TIGHTNESS. (b) CBRE NA asking lease-rate YoY growth +18.6% → +7.0% → +12.6% → +6.5% (`lease_rate_growth_2025_6_5pct`; `cbre-na-asking-lease-rate`, 2023-H2 to 2025, $n = 4$, grade B): MIXED (level still rising, growth decelerating; growth rates from ANALYSIS §5). (c) Prior-gen GPU cheapest-listed 30-day deltas: A100 −19.0%, L40S −33.2% (`priorgen_gpu_30d_a100_-19_l40s_-33`; `mlq-gpu-30d-delta` / `marketplace-a100-floor` / `marketplace-l40s-floor`, 2026-09, grade C: labeled as context, not a headline): SOFTNESS (prior-gen only). (d) H100 marketplace 24h price dispersion 21× ($0.72–$15.14/hr, 141k+ obs; `h100_marketplace_dispersion_21x`, `aimc-h100-dispersion`, 2026-05-13, grade B): SOFTNESS/fragmentation. (e) Microsoft lease cancellations/deferrals ∼2 GW, US+Europe (`msft_lease_cancellation_2gw`; events_pricing.csv Microsoft event, 2025-03-26, grade B*): SOFTNESS (demand-side). (f) JLL NA preleased share 73% (2025-H1) → 92% (2025-YE; `preleased_2025ye_92pct`, `jll-na-preleased`, grade B) with CBRE NA net absorption 1,809.5 → 2,497.6 MW (`absorption_2025_2498mw`, `cbre-na-net-absorption`, grade B); 9 GW delivered in 2025, 35 GW under construction: TIGHTNESS. Net read: near-term physical market still tight, but every price signal at the margin points to softening.

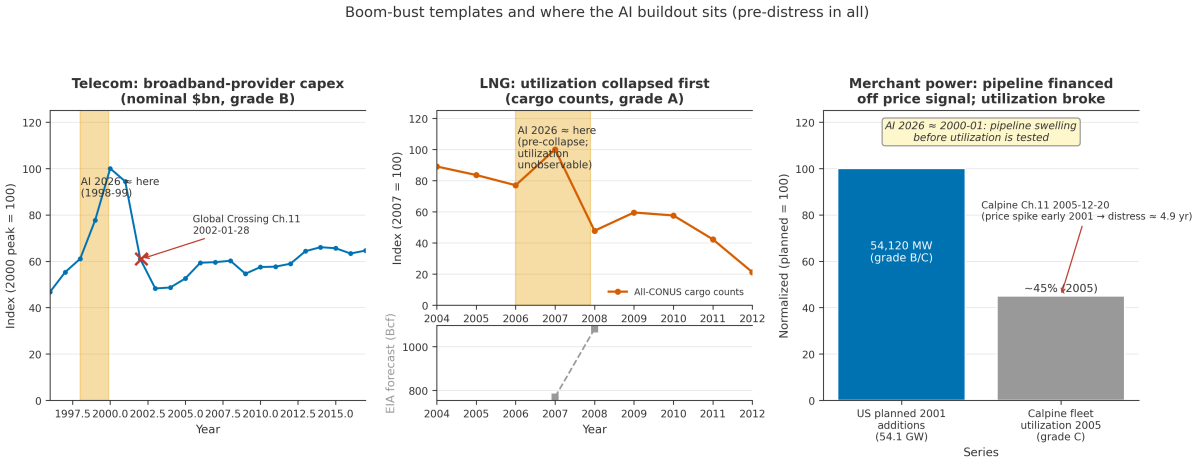


Figure 7: Boom-bust templates and where the AI buildout sits (pre-distress in all), each normalized. Left: telecom broadband-provider capex 1996–2017 (`telecom_capex_broadband_providers`, nominal $bn, grade B), indexed to the 2000 peak ($118.1bn = 100), trough 2003 ($57.0bn); red X marks Global Crossing Ch. 11 (2002-01-28; `telecom_distress_bankruptcy`, grade B); yellow band = AI 2026 ≈ telecom 1998–99 (margin price softness while the build accelerates; price-to-distress lead ≈ 1.9 yr, `analogue_telecom_price_to_distress_lead_yrs`). Middle: LNG all-CONUS import cargo counts 2004–2012 (`lng_cargos_by_terminal`, grade A), indexed to 2007 (274 = 100): 274 → 131 (2008) → 58.3 (2012), a 78.7% collapse (`analogue_lng_cargo_collapse_2007_2012`); gray dashed = EIA Jan-2007 forecast volumes 770/1,080 Bcf (`lng_import_volumes_forecast`, grade A); expectations rose as utilization collapsed; yellow band = AI 2026 ≈ pre-collapse, with the caveat that AI-fleet utilization is currently unobservable. Right: merchant power. US planned 2001 additions 54,120 MW (`merchant_power_pipeline_planned_us_total`, 2001, grade B/C) vs. Calpine fleet utilization ~45% in 2005 (`merchant_power_utilization_calpine_fleet`, grade C, labeled); Calpine Ch. 11 2005-12-20 (`merchant_power_distress_bankruptcy`); price spike (early 2001) → distress ≈ 4.9 yr (`analogue_merchant_power_price_to_distress_lead_yrs`). AI 2026 ≈ 2000–01: pipeline swelling off the price signal before utilization is tested. Caveat: analogue monetary figures are nominal as published and never mixed with the nominal capex series.

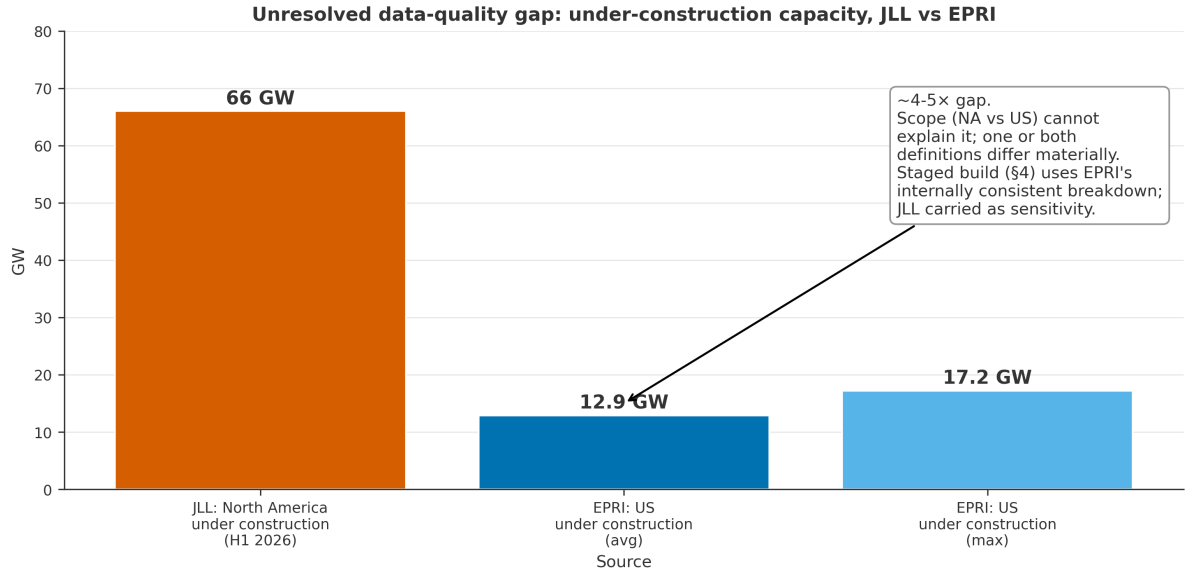


Figure 8: The unresolved inconsistency, shown honestly: a data-quality figure, not a result. JLL reports 66 GW under construction in North America (H1 2026; `jll_na_under_construction`, grade B, catalog-transcribed) vs. EPRI's US under-construction avg 12.9 GW / max 17.2 GW (`epri_us_capacity_pipeline_under_construction_avg/max`, grade A, catalog-transcribed): a ∼4–5× gap that scope (NA vs US) cannot explain; one or both definitions differ materially. The staged 2030 supply build (Figure 4) uses EPRI's internally consistent stage breakdown and carries the JLL figure as a sensitivity input. Both values transcribed from catalog; verify from primary (JLL H1 2026 report; EPRI 2026) before use. No dedicated `key_estimates.csv` rows exist for these levels (see FIGURE_LOG.md).